

# Environmental data

## Top 10 genus

library prep

install

activate

```
library(readxl)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
library(ggplot2)
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

```
date, intersect, setdiff, union
```

```
library(patchwork)
```

## Retrieve data

### Home PC

```
#metadata
setwd("C:/Results")

metadata <- read_excel("samples Oslo Fjord.xlsx")
```

### School PC

## Data cleaning

```
#Keep only rows with the station IDs i need and all negatives
metadata <- subset(metadata, station_id %in% c("R-5", "S-9", "Negative"))

#remove the negatives i dont need
metadata <- subset(metadata, !(file_name_R1 %in% c("14-OS24_S14_L001_R1_001.fastq.gz", "21-OS24_S14_L001_R1_001.fastq.gz")))

#getting rid of unnecessary columns
metadata_clean <- metadata %>%
  select(-c(
    DNA_RNA,
    fraction_name_original,
    fraction_name,
    fraction_max,
    fraction_min,
    project,
    cruise,
    station_set,
    station_id_num,
    time,
    season,
```

```

    day_length,
    depth_level,
    depth,
    elevation,
    depth_range,
    substrate,
    latitude,
    longitude,
    site_name,
    country,
    oceanic_region,
    ecosystem,
    Qubit
  ))

plot_data <- metadata_clean %>%
  filter(station_id %in% c("R-5", "S-9"))

```

## ploting

```

plot_data <- plot_data %>%
  mutate(
    nitrate_nitrite = as.numeric(nitrate_nitrite),
    ammonia = as.numeric(ammonia),
    total_nitrogen = as.numeric(total_nitrogen),
    temperature = as.numeric(temperature) # if not numeric already
  )

```

Warning: There was 1 warning in `mutate()`.

i In argument: `ammonia = as.numeric(ammonia)`.

Caused by warning:

! NAs introduced by coercion

```

plot_data <- plot_data %>%
  filter(ammonia != "<3")

# Remove "<3" from ammonia and convert all relevant columns to numeric
plot_data <- plot_data %>%
  filter(ammonia != "<3") %>%

```

```

mutate(
  temperature = as.numeric(temperature),
  nitrate_nitrite = as.numeric(nitrate_nitrite),
  ammonia = as.numeric(ammonia),
  total_nitrogen = as.numeric(total_nitrogen)
)

# Helper function to create a line plot
plot_line_graph <- function(data, y_var, y_label, title) {

  # Compute 10 nice round numbers for y-axis
  y_breaks <- pretty(data[[y_var]], n = 10)

  ggplot(data, aes_string(x = "date", y = y_var, color = "station_id", group = "station_id")) +
    geom_line(linewidth = 1) +
    geom_hline(yintercept = y_breaks, color = "grey80") +
    scale_x_date(date_breaks = "1 month", date_labels = "%b %y") +
    scale_y_continuous(breaks = y_breaks) +
    labs(
      title = title,
      x = "Date",
      y = y_label,
      color = "Station"
    ) +
    theme_minimal() +
    theme(
      panel.grid = element_blank(), # remove default grid
      axis.text.x = element_text(angle = 45, hjust = 1)
    )
}

# Temperature plot (BOC)
temp_plot <- plot_line_graph(plot_data, "temperature", "Temperature (°C)", "Temperature")

```

Warning: `aes\_string()` was deprecated in ggplot2 3.0.0.

i Please use tidy evaluation idioms with `aes()`.

i See also `vignette("ggplot2-in-packages")` for more information.

```

# Nitrate + Nitrite plot (mg/L)
nitrate_plot <- plot_line_graph(plot_data, "nitrate_nitrite", "Nitrate + Nitrite (µg/L)",

```

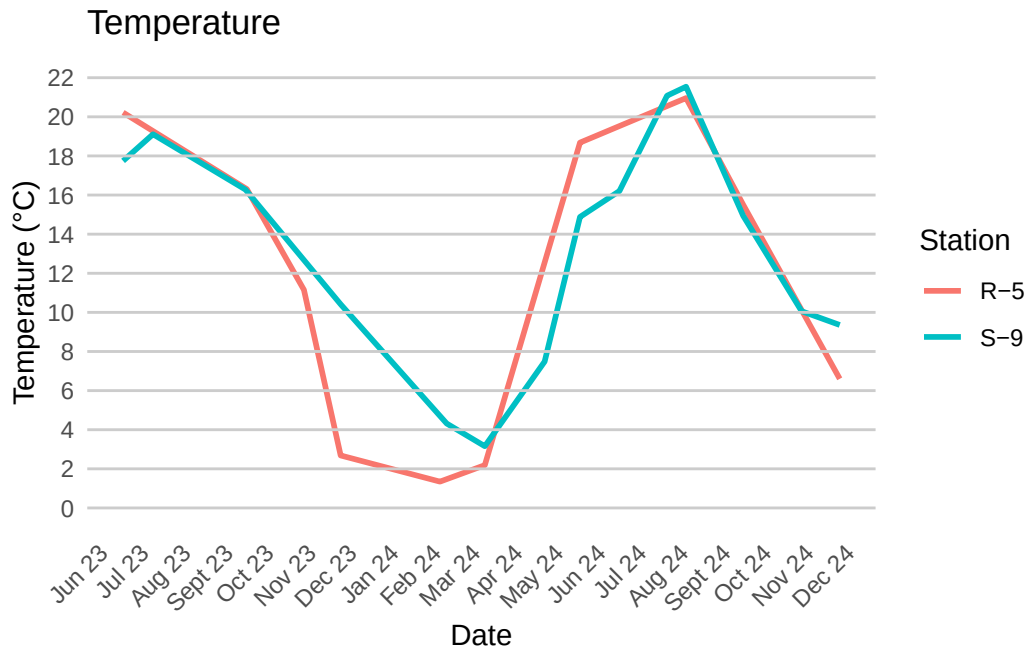
```

# Ammonia plot (mg/L)
ammonia_plot <- plot_line_graph(plot_data, "ammonia", "Ammonia (µg/L)", "Ammonia")

# Total Nitrogen plot (mg/L)
totalN_plot <- plot_line_graph(plot_data, "total_nitrogen", "Total Nitrogen (µg/L)", "Total Nitrogen")

# Display plots
temp_plot

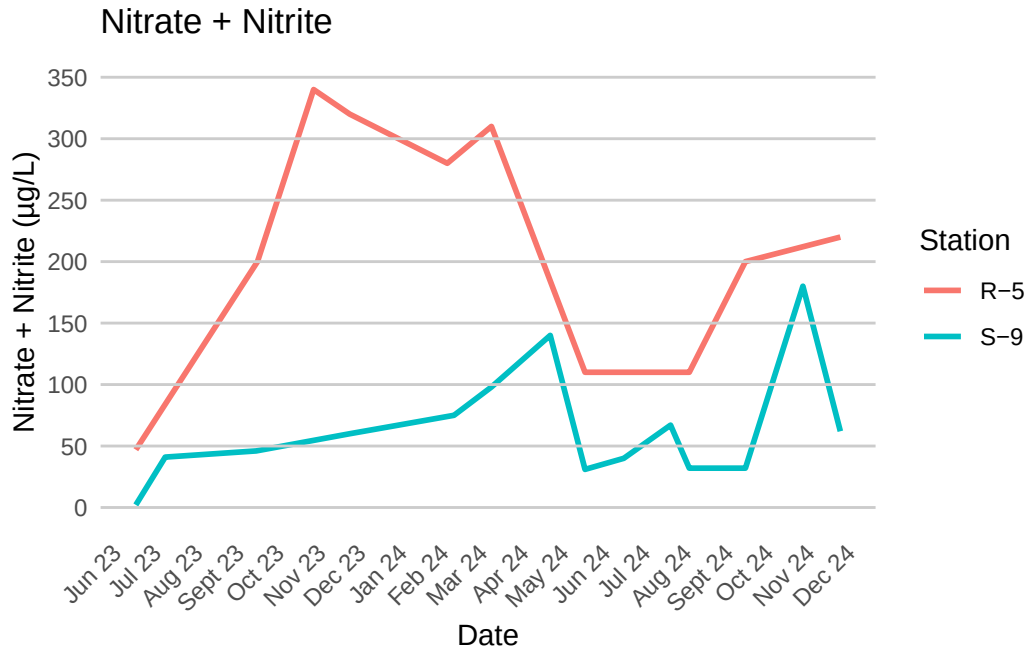
```



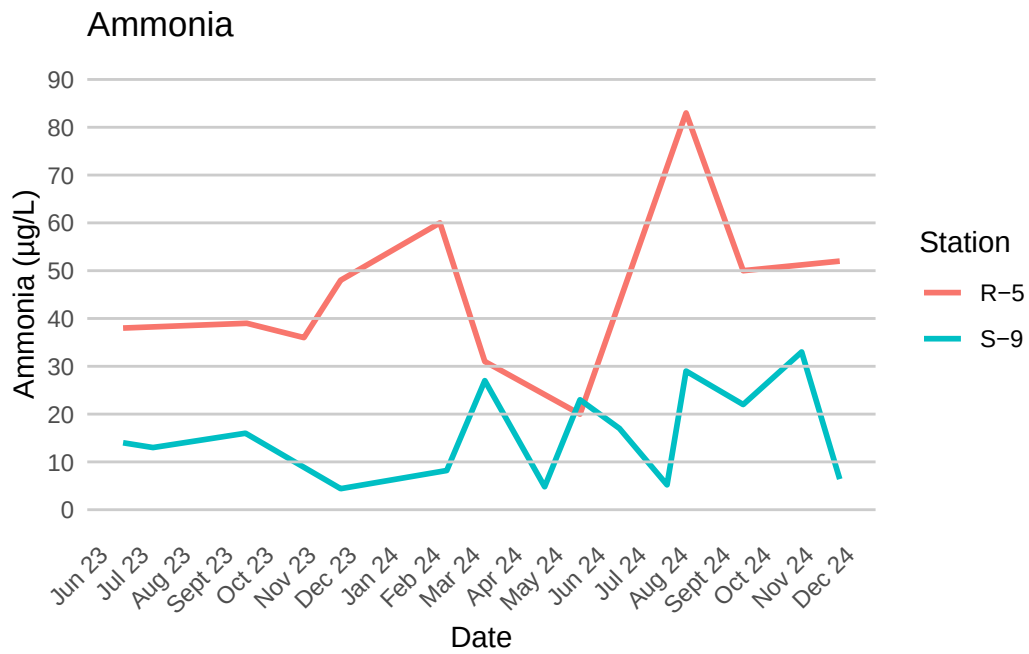
```

nitrate_plot

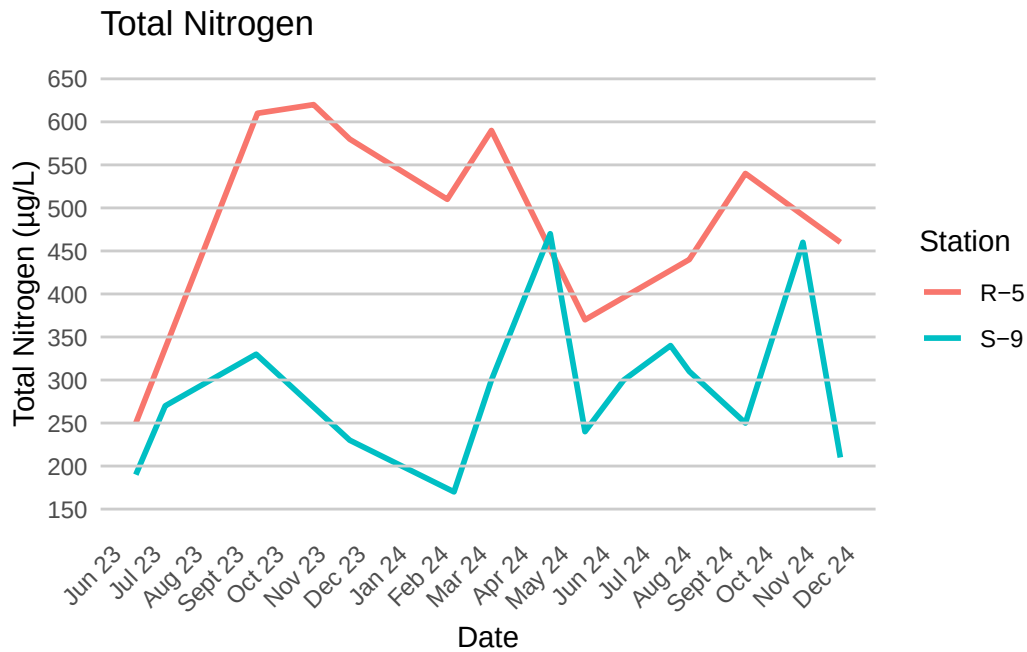
```



ammonia\_plot



```
totalN_plot
```



```
# Convert columns to numeric
plot_data <- plot_data %>%
  mutate(
    fluorescence = as.numeric(fluorescens),
    secchi_depth = as.numeric(secchi_depth),
    silicate = as.numeric(silicate),
    salinity = as.numeric(salinity)
  )
```

Warning: There was 1 warning in `mutate()`.

i In argument: `secchi\_depth = as.numeric(secchi\_depth)`.

Caused by warning:

! NAs introduced by coercion

```
# Helper function (same as before)
plot_line_graph <- function(data, y_var, y_label, title) {
  y_breaks <- pretty(data[[y_var]], n = 10) # 10 round numbers
```

```

ggplot(data, aes_string(x = "date", y = y_var, color = "station_id", group = "station_id")) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = y_breaks, color = "grey80") +
  scale_x_date(date_breaks = "1 month", date_labels = "%b %y") +
  scale_y_continuous(breaks = y_breaks) +
  labs(
    title = title,
    x = "Date",
    y = y_label,
    color = "Station"
  ) +
  theme_minimal() +
  theme(
    panel.grid = element_blank(),          # no default grid
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
}

# Fluorescence plot (AU)
fluorescence_plot <- plot_line_graph(plot_data, "fluorescence", "Fluorescens (AU)", "Chlorophyll")

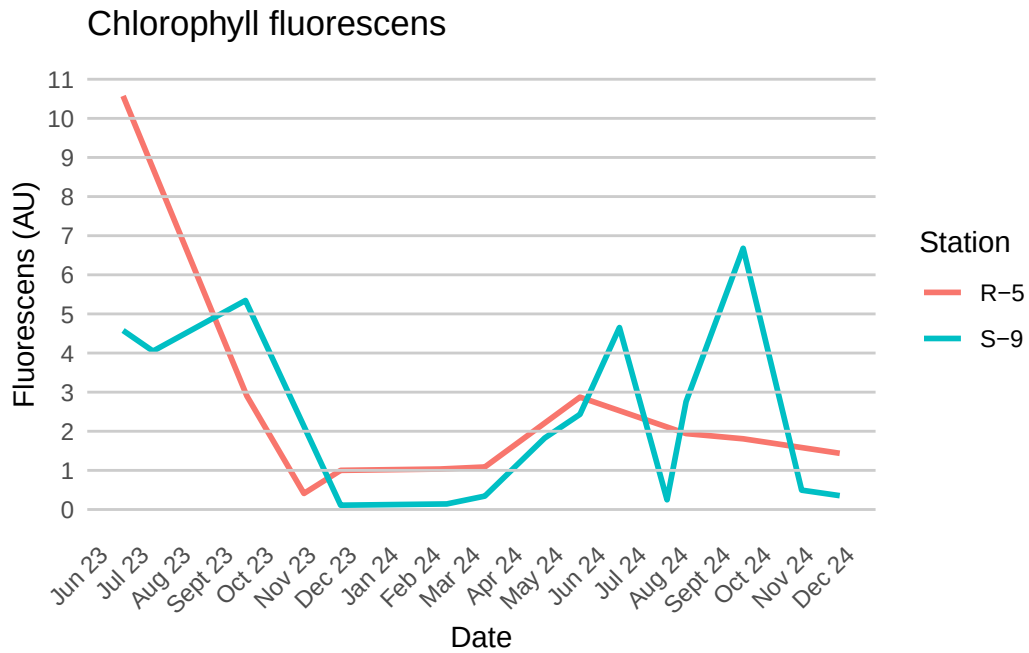
# Secchi depth plot (m)

# Silicate plot (mg/L)
silicate_plot <- plot_line_graph(plot_data, "silicate", "Silicate (mg/L)", "Silicate")

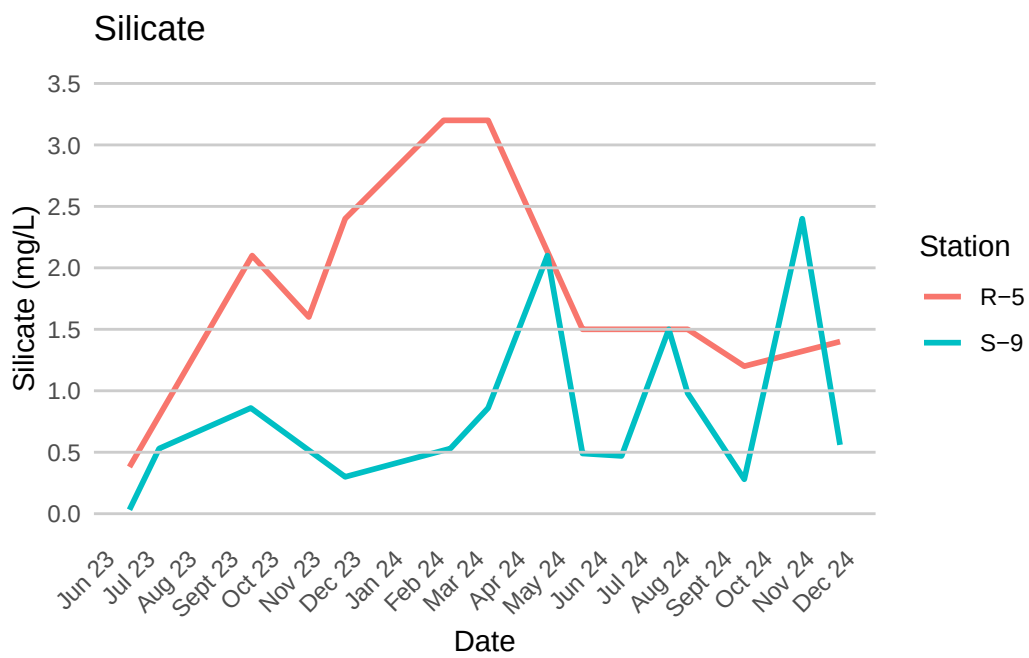
# Salinity plot (PSU)
salinity_plot <- plot_line_graph(plot_data, "salinity", "Salinity (PSU)", "Salinity")

# Display plots
fluorescence_plot

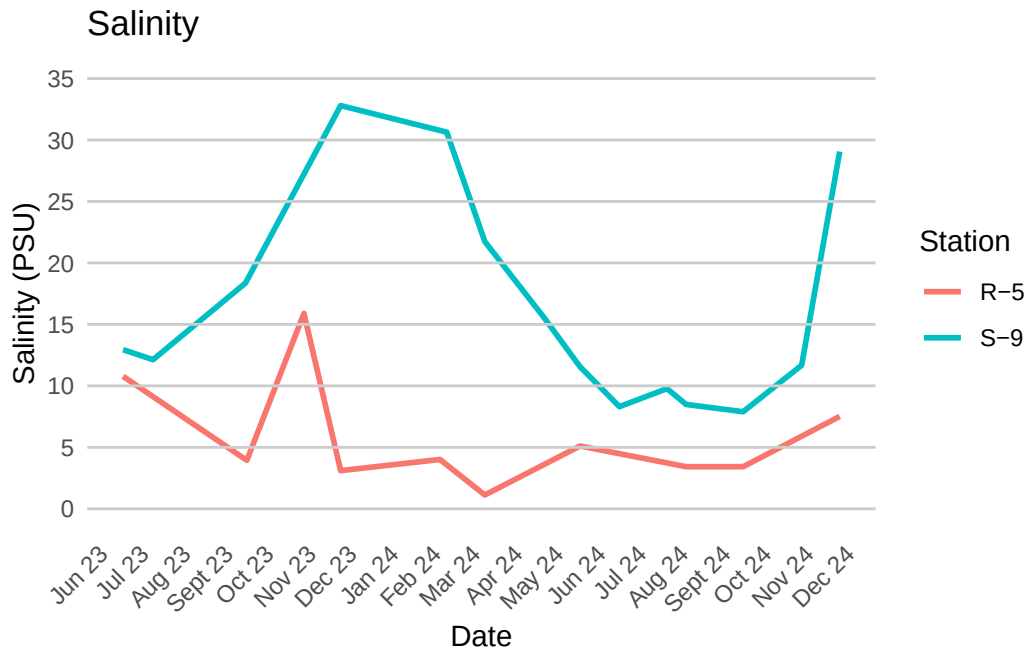
```



`silicate_plot`



```
salinity_plot
```



```
plot_data_secchi <- plot_data %>%
  filter(!is.na(secchi_depth))

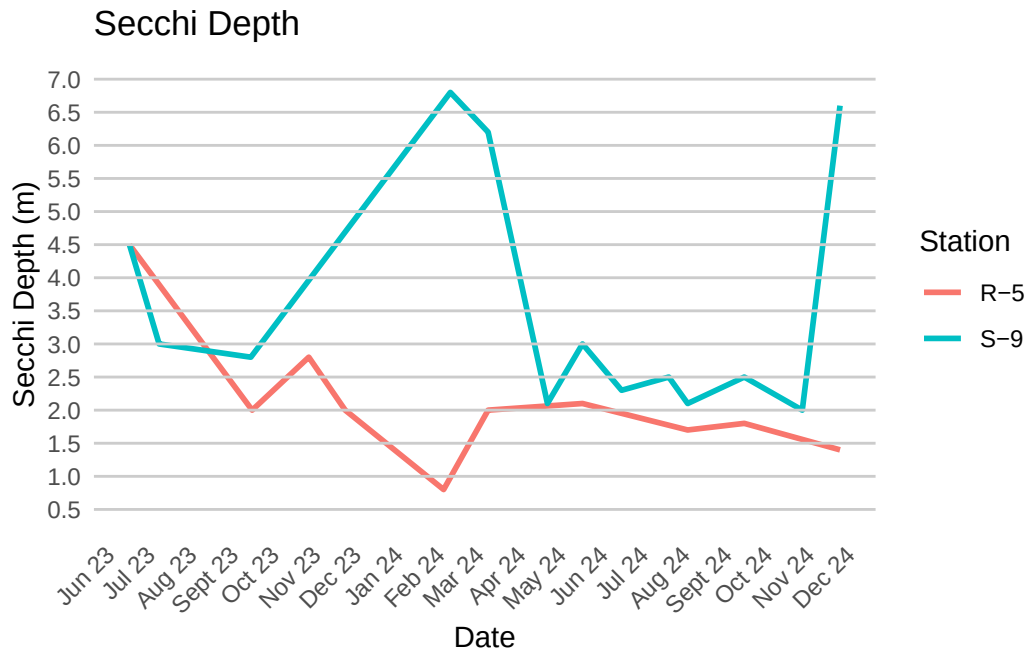
# 10 round numbers for y-axis
y_breaks <- pretty(plot_data_secchi$secchi_depth, n = 10)

secchi_plot <- ggplot(plot_data_secchi, aes(x = date, y = secchi_depth, color = station_id)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = y_breaks, color = "grey80") +
  scale_x_date(date_breaks = "1 month", date_labels = "%b %y") +
  scale_y_continuous(breaks = y_breaks) +
  labs(
    title = "Secchi Depth",
    x = "Date",
    y = "Secchi Depth (m)",
    color = "Station"
  ) +
  theme_minimal() +
  theme(
```

```

    panel.grid = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
secchi_plot

```



## combined

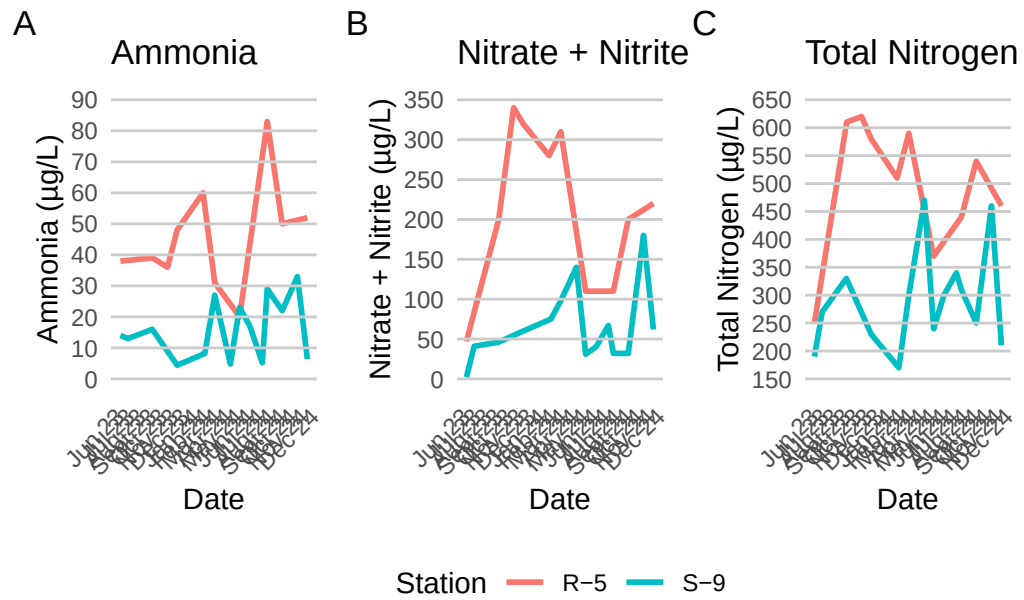
```

library(patchwork)

nutrient_grid <- ammonia_plot + nitrate_plot + totalN_plot +
  plot_layout(ncol = 3, guides = "collect") + # collect shared legend
  plot_annotation(tag_levels = "A") &
  theme(legend.position = "bottom")           # place legend at the bottom

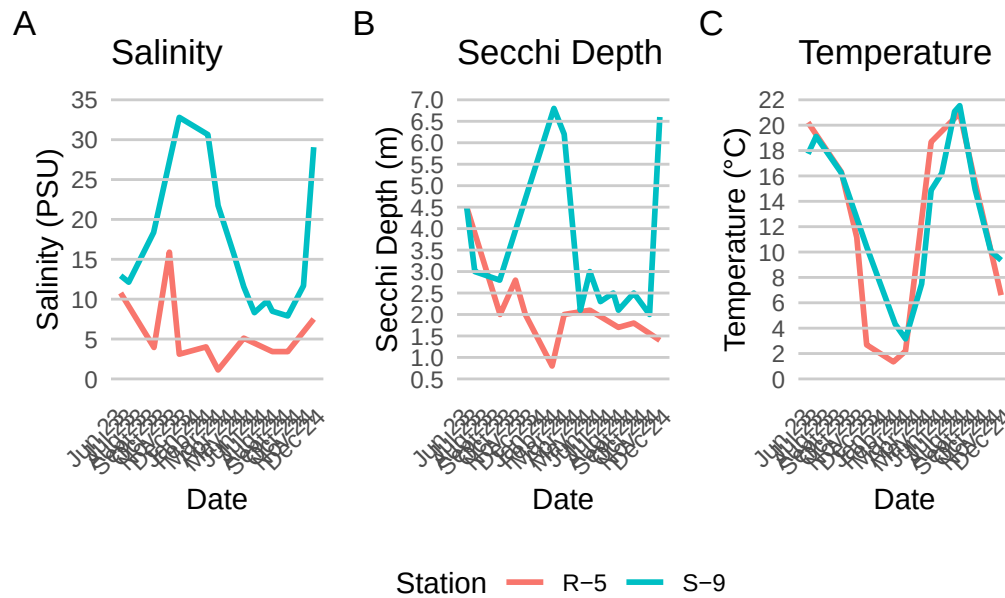
nutrient_grid

```



```
env_grid <- (salinity_plot + secchi_plot + temp_plot) +
  plot_layout(guides = "collect") +
  plot_annotation(tag_levels = "A") &
  theme(legend.position = "bottom")

env_grid
```



```
# Convert to numeric if needed
plot_data <- plot_data %>%
  mutate(phosphate = as.numeric(phosphate))
```

Warning: There was 1 warning in `mutate()`.  
 i In argument: `phosphate = as.numeric(phosphate)`.  
 Caused by warning:  
 ! NAs introduced by coercion

```
# Create 10 round y-axis values
y_breaks_phos <- pretty(plot_data$phosphate, n = 10)

phosphate_plot <- ggplot(plot_data, aes(x = date, y = phosphate, color = station_id, group = station_id)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = y_breaks_phos, color = "grey80") +
  scale_x_date(date_breaks = "1 month", date_labels = "%b %y") +
  scale_y_continuous(breaks = y_breaks_phos) +
  labs(
    title = "Phosphate",
    x = "Date",
    y = "Phosphate (µg/L)",
```

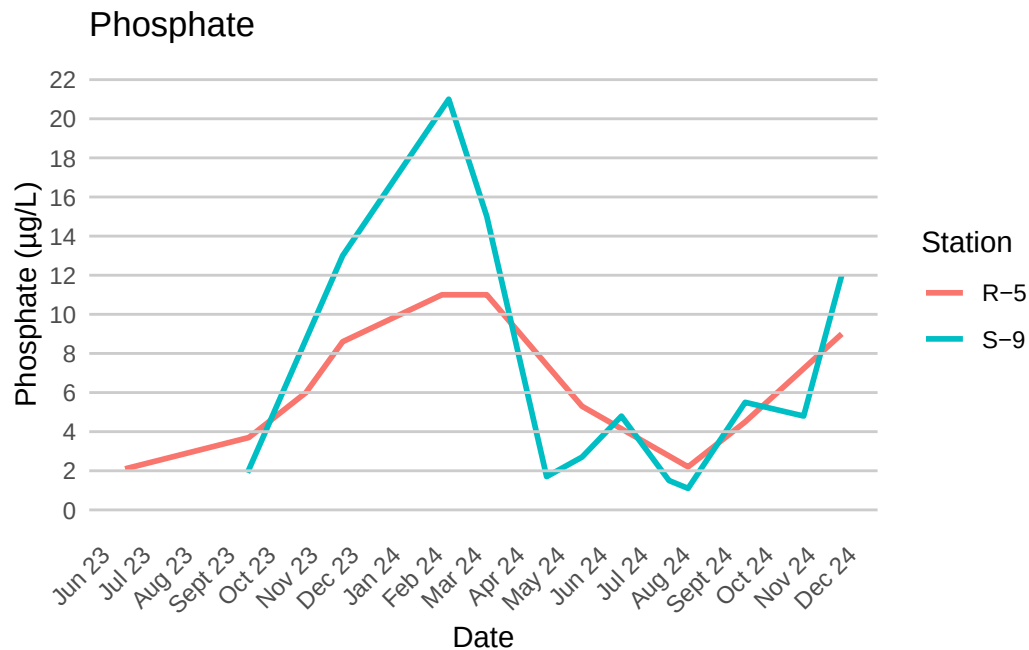
```

    color = "Station"
  ) +
  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

```

phosphate\_plot

Warning: Removed 2 rows containing missing values or values outside the scale range (`geom\_line()`).



```

combined_psf <- phosphate_plot + fluorescence_plot + silicate_plot +
  plot_layout(ncol = 3, guides = "collect") + # 3 side-by-side
  plot_annotation(tag_levels = "A") & # A, B, C labels
  theme(legend.position = "bottom") # one shared legend

```

combined\_psf

Warning: Removed 2 rows containing missing values or values outside the scale range (`geom_line()`).

